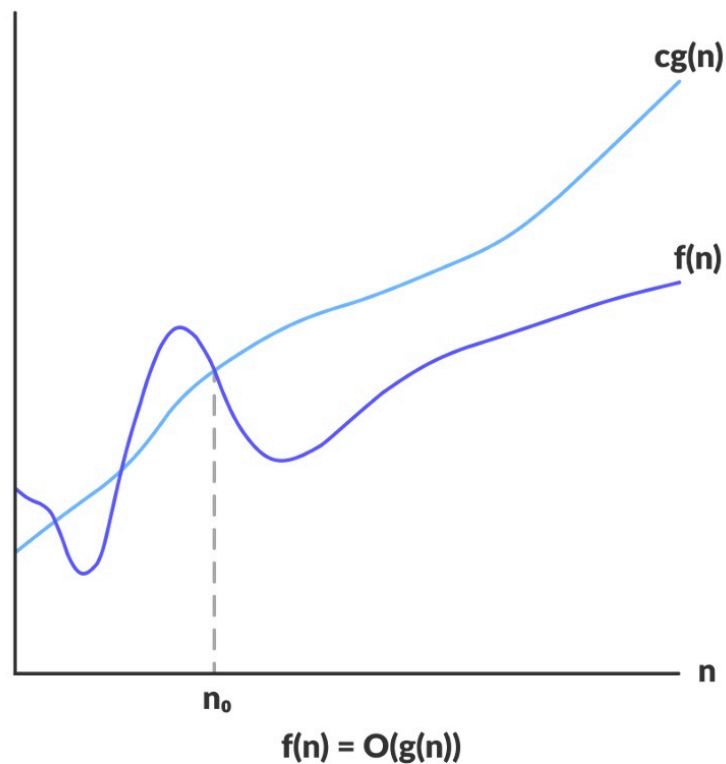


Big-O Notation

$O(g(n)) = \{f(n) : \text{there exists +ve constants } n_0 \text{ \& } c \text{ such that } 0 \leq |f(n)| \leq c|g(n)| \text{ for all } n \geq n_0\}$



Example,

$$f(n) = 3n^2 + 5n + 7 \quad g(n) = n^2$$

$$f(n) \in O(g(n)) \iff \exists c > 0, n_0 \text{ such that } 0 \leq f(n) \leq cg(n) \quad \forall n \geq n_0$$

We make a hypothesis:

$$5n^2 \leq 5n \quad \Delta \quad 7n^2 \leq 7 \quad , \quad n \geq 1$$

We replace the hypothesis in $f(n)$. We can do this because it will not change the dominant term, which is still n^2

For $n \geq 1$,

$$3n^2 + 7n^2 + 5n^2 \leq c \cdot n^2$$

$$15n^2 \leq c n^2$$

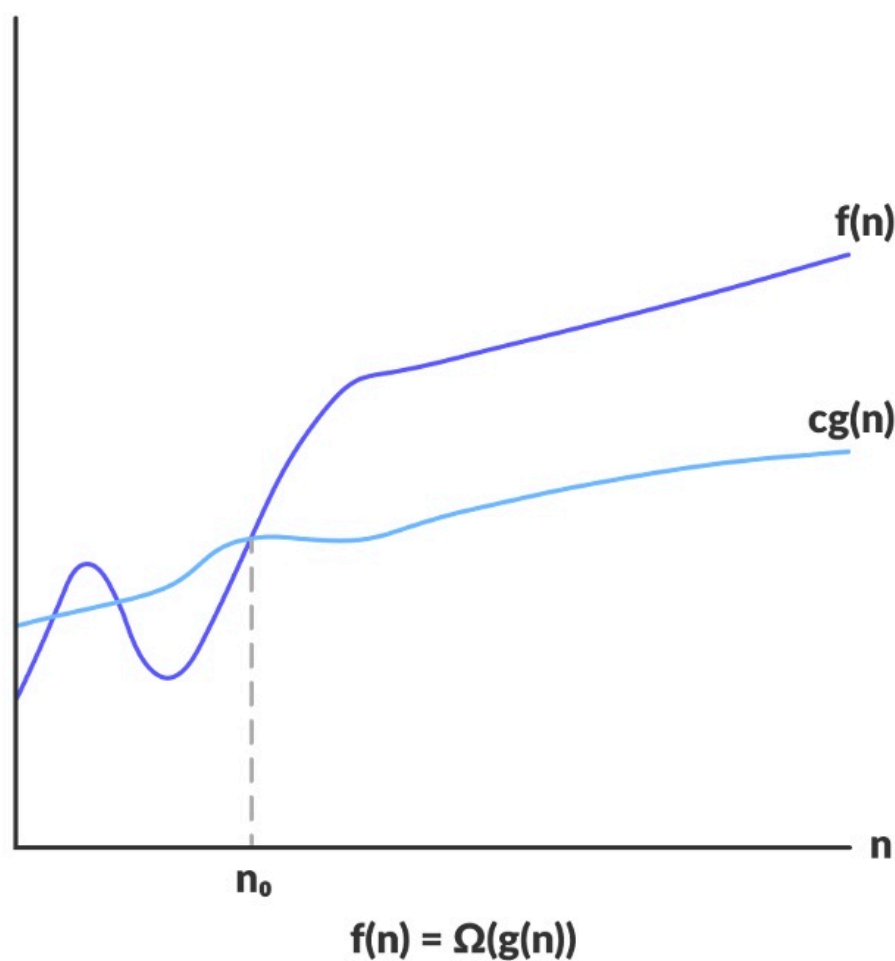
$$c \geq 15$$

$$n_0 = 1$$

Thus, $f(n) \in O(n^2)$

Big- Ω Notation

$\Omega(g(n)) = \{f(n) : \text{there exists +ve constants } n_0 \text{ \& } c \text{ such that } 0 \leq c|g(n)| \leq |f(n)| \text{ for all } n \geq n_0\}$



Example,

$$f(n) = 3n^2 + 5n + 7 \quad g(n) = n^2$$

$$3n^2 + 5n + 7 \geq c \cdot n^2 \quad \forall n \geq 1$$

Dominant terms constant as c or less

So,

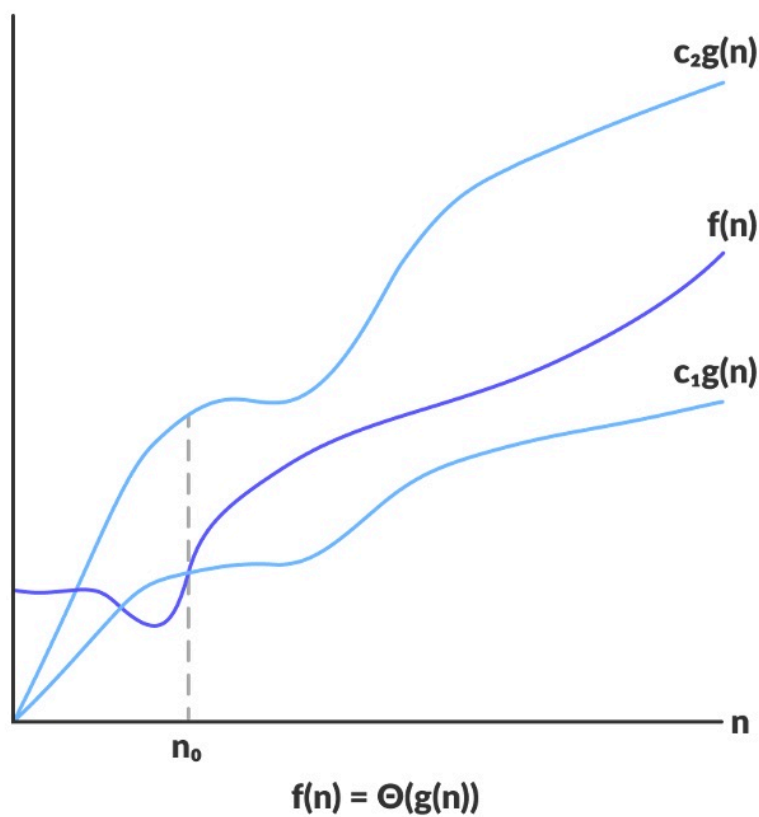
$$c \geq 3$$

$$n_0 = 1$$

Thus, $f(n) \in \Omega(n^2)$

Theta Notation

$$O(g(n)) = \left\{ f(n) : \text{there exists +ve constants } n_0, c_1, \text{ \& } c_2 \text{ such that } 0 \leq c_1 g(n) \leq f(n) \leq c_2 g(n) \text{ for all } n \geq n_0 \right\}$$



Example,

$$f(n) = 3n^2 + 5n + 7 \quad g(n) = n^2$$

From above,

$$3n^2 \leq 3n^2 + 5n + 7 \leq 15n^2$$

So,

$$n_0 = 1$$

$$c_1 = 3$$

$$c_2 = 15$$

Thus,

$$f(n) \in \Theta(n^2)$$

NOTE: Big-Theta exists if big-O & big Ω are same i.e.

$$O(g(n)) = \Omega(g(n)) \Rightarrow \Theta(g(n))$$

Example,

$$f(n) = 4n \log n + 10n \quad g(n) = n \log n$$

$$4n \log n + 10n \leq c \cdot n \log n$$

We make hypothesis,

$$10n \leq 10n \log n \quad \text{for } n \geq 2$$

$$28 \leq 8$$

Now,

$$4n \log n + 10n \log n \leq c \cdot n \log n$$

$$14n \log n \leq c \cdot n \log n$$

So,

$$c = 14$$

$$n_0 = 2$$

Thus,

$$f(n) \in O(n \log n)$$

Now,

$$4n \log n + 10n \geq c \cdot n \log n$$

So,

$$c = 4$$

$$n_0 = 2$$

Thus,

$$f(n) \in \Omega(n \log n)$$

Lastly,

$$0 \leq 4n \log n \leq 4n \log n + 10n \leq 14n \log n$$

So,

$$c_1 = 4$$

$$c_2 = 14$$

$$n_0 = 2$$

Thus,

$$f(n) \in \Theta(n \log n)$$

Exercise:

2.2.1

$$f(n) = \frac{n^3}{1000} + 100n^2 - 100n + 3$$

We compare against $g(n) = n^3$

We make hypothesis;

$$100n^2 \leq 100n^3, \quad 100n \leq 100n^3, \quad 3 \leq 3n^3 \quad \text{for } n \geq 1$$

Now,

$$\frac{n^3}{1000} + 100n^3 + 100n^3 + 3n^3 \leq c \cdot n^3$$

$$n^3 \left(\frac{1}{1000} + 100 + 100 + 3 \right) \leq c \cdot n^3$$

$$203.001 n^3 \leq c \cdot n^3$$

So,

$$c = 203.001$$

$$n_0 = 1$$

For big Ω ;

$$c = \frac{1}{1000}$$

$$n_0 = 1$$

For big Θ ;

$$0 \leq \frac{1}{1000} n^3 \leq \frac{n^3}{1000} + 100n^2 - 100n + 3 \leq 203.001 n^3$$

$$c_1 = \frac{1}{1000}$$

$$n_0 = 1$$

$$c_2 = 203.001$$

